FACTORS AFFECTING ROAD SAFETY IN THE UNITED STATES: A DEEP DIVE INTO NHTSA GES 2015 FILES*

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***Abstract*—This paper investigates the factors influencing road safety in the United States using the National Highway Traffic Safety Administration's General Estimates System (NHTSA GES) 2015 dataset. Despite advancements in vehicle safety technology and infrastructure improvements, high rates of traffic accidents persist. The study explores the implications of driver behavior, vehicle attributes, environmental conditions, and infrastructural deficiencies in shaping crash outcomes. Through the application of machine learning models, notably Random Forest, we assess injury severity based on vehicle characteristics. Key findings reveal significant correlations between vehicle age, weight, and size with crash severity, highlighting the need for targeted interventions. Recommendations underscore the importance of public safety campaigns, legislative measures for vehicle safety, and infrastructure enhancements. The research concludes with a call for future studies that integrate real-time data analytics and examine the implications of automation in road safety.**

***Index Terms*—Road safety, traffic accidents, NHTSA, machine learning, vehicle characteristics, driver behavior, environmental factors, predictive modeling, public policy**

I. Introduction

A. Background

The problem of road safety in the United States is still significant even though there have been numerous developments in the hardware of both automobiles and roads in the past few decades. In its reports, the National Highway Traffic Safety Administration (NHTSA)'s report revealed that motor vehicle crashes still cause over 38,000 deaths per year and many non-fatal injuries. This is a cause for concern because the U.S. has invested a lot of money in making vehicles safer; from air bags, anti-locking brakes, electronic stability control, to advance driving assistance systems. In addition, much quantity has been invested in the development of roads in the country, redesigning of highways, and launching various traffic safety awareness campaigns. However, the rates of motor vehicle crashes, injuries, and fatalities continue a high index indicating the overall difficulty in addressing needs identification and applying effective countermeasures, which requires a better understanding of the nature of the road safety issue. However, progress in vehicle technologies has undoubtedly helped improve road safety; still, the effective approach that acts on all aspects of exposure consisting of vehicle attributes, driver behavior, environmental conditions, and regulatory measures is needed. This paper discusses contributing factors like distracted driving, impaired driving, speeding and condition of infrastructure of the roads in the calculation of the rate of growth of frequency and severity of accidents. In addition, interactions between the vehicle properties including age, size, and safety measures and corresponding driving actions and the condition of the environment increases the difficulty of reducing accidents. An even deeper insight into certain processes at work can be gained by examining greater numbers of traffic accidents files including NHTSA GES files which contains rich data about the circumstances of crashes in the United States. What is more, this dataset that contains vehicle characteristics, driver behavior, crash severity and environmental factors will allow to find the relationships and patterns that cannot be noticed when using simple analysis tools.

B. Problem Statement

Even with the ongoing improvements in vehicle safety and infrastructure, there are still numerous traffic accidents, injuries, and fatalities in the United States. Not only the actions of the driver, but also vehicle characteristics, environmental conditions and infrastructure design are among the more important factors that lead to these types of incidents. It is important to understand how the various characteristics of vehicles (such as their size, weight, safety technologies) affect crash severity and occurrences. Addressing the root causes of road safety issues can be achieved by examining how external factors such as road conditions, weather, and traffic patterns impact vehicle types and driver behavior. This study seeks to address the gap by examining the NHTSA GES 2015 dataset and investigate how vehicle characteristics can impact crash results. This increased understanding will inform targeted actions to reduce accidents and deaths, as well as help develop evidence-based safety policies. This research employs advanced analytical methods, such as machine learning, to identify areas where traditional statistical methods may not

be effective and potentially contribute to the improvement of road safety in America.

C. Objectives

The main objective of this study is to explore the relationship between vehicle characteristics and road safety outcomes, with a focus on crash severity. Specifically, the research aims to: Identify the key vehicle characteristics (e.g., make, model, year, body type, safety features) that contribute most significantly to crash severity.

- Examine the influence of environmental factors (weather, road surface conditions, visibility) on the frequency and severity of accidents involving different vehicle types.
- Analyze driver behavior in conjunction with vehicle attributes to understand the human factors contributing to traffic accidents.
- Develop a predictive model using machine learning techniques to estimate crash severity based on vehicle and environmental features.

D. Research Questions

To guide the investigation, the following research questions will be addressed:

E. General Research Question

- What is the relationship between vehicle characteristics (make, model, year, body type) and the severity of crash outcomes?
- How do specific vehicle attributes, such as safety features (e.g., airbags, ABS), weight, and size, influence the likelihood and severity of crashes?
- What patterns can be identified in crash occurrences based on vehicle types (e.g., passenger cars, trucks, SUVs) and their characteristics?
- How does the age of a vehicle correlate with the frequency and severity of accidents?

F. Exploratory data analysis question

- What is the relationship between vehicle characteristics (make, model, year, body type) and the severity of crash outcomes?
- How do specific vehicle attributes (e.g., safety features, weight, size) influence the likelihood and severity of collisions?
- What patterns can be identified in crash occurrences based on vehicle types involved (e.g., passenger cars vs. trucks) and their characteristics?
- How does the age of a vehicle correlate with the frequency and severity of accidents?

G. Machine learning questions

- Can machine learning models effectively predict crash severity based on vehicle characteristics compared to traditional statistical methods?
- How effective are deep learning models in identifying relationships between vehicle features and crash outcomes from historical accident data?
- What role can supervise learning techniques play in uncovering hidden patterns within vehicle crash data related to specific vehicle attributes?
- How can clustering algorithms be utilized to group similar vehicles based on accident involvement and identify at-risk categories?
- What features derived from vehicle characteristics contribute most significantly to predicting accident severity using machine learning approaches?

H. Motivation

The rationale for this study is to feed into the existing literature with a view of reducing deaths and injuries caused by traffic accidents in the United States. It is with the hope of establishing which aspects of vehicles contribute most to the crash severity that this study would want to draw attention to policy makers and potential vehicle safety legislation that can help save lives. Besides, as a result of using machine learning models, this paper will identify innovative data-driven solutions for reducing road traffic accidents that prevent them before happening. Another significant factor is related to the identification of road safety precursors that fit not only in the context of engineering strategies for vehicle and road construction, or in addressing public awareness problems but also in the development of additional regulations. If the current study is meant to reveal the truth and needs to make an impact on policy and practice, then the following are the ways that the observation would best impact: In this study, therefore, the authors endeavored to examine the relationships between and among vehicle attributes, driver behavior and environmental factors in the hope that the findings will be useful in deriving solutions to lessen road traffic fatalities and injuries in the USA.

II. Literature Review

A. Introduction

[1] argue that road safety in the United States of America is determined by a combination of factors that range from behavioral, environmental, infrastructural, and technological. This is because the United States of America is a large and diverse country presenting different driving conditions from various state laws, further complicating road safety. In so doing, road safety in the U.S. is not yet guaranteed compared to other developed nations, with the National Highways and Traffic Safety Administration (NHTSA) recording more than 38,000 fatalities from motor vehicle crashes. These deafening poor road safety conditions are primarily attributed to speeding, impaired driving, and the vulnerability of pedestrians and cyclists. [2] connote the dangers associated with distracted driving due to the desire to use smartphones and in-vehicle technologies. Also, ageing road infrastructure has primarily contributed to the safety of road users, especially the lack of capacity and infrastructural design that can accommodate modern traffic volumes. Adverse weather conditions still threaten road safety, and many motor vehicle crashes have been linked to weather conditions that distract road users.

Despite evidence of persistent poor road safety conditions for road users, there has been an improvement, especially with technological advancements in vehicles like automatic emergency braking and lane departure warnings that have helped reduce crashes significantly.

B. The Safe System Approach

It is a road safety framework rooted in the understanding that human beings, who are the primary road users, make mistakes. A holistic approach is required to account for human error and create a safer transportation system. This is to help reduce traffic-related deaths and injuries by designing safer roads, vehicles, speeds and policies that ensure all road users' safety (Job et al., 2022). [3] The approach considers vital principles, such as human beings making mistakes and therefore seeking to minimize the physical vulnerability to a crush. It also seeks to vest the responsibility of road safety in all road users through safer roads, management of safe speeds, and safe vehicles by educating road users on some of these critical principles. The approach outlines a rapid and effective post-crash response to reduce the fatalities associated with traffic crashes. Booker et al. (2024) [4] comment on applying the safe system approach in the U.S., especially vision zero initiatives that aim to eliminate all traffic fatalities and injuries. The zero vision strategies involve policies seeking to reduce speed limits, redesign streets, and improve infrastructure for pedestrians and cyclists. The safe system framework has been acknowledged due to its recognition of the inevitability of human error and the quest to design roads, vehicles, and policies to alleviate the consequences of human error.

C. Influence of Driver Behavior on Road Safety

[5] Farooq et al. (2020) underscore that driver behavior has significantly influenced the safety of all road users in the United States. Traffic crashes are not solely caused by human error but also by road infrastructure, vehicle safety features and traffic laws. According to reports, the use of mobile phones during driving is responsible for almost 33 percent of traffic collisions. 142. The use of alcohol and drugs in driving caused almost 30 percent of all fatalities. Furthermore, speeding and aggressive driving were responsible for almost 29 percent of road accident fatalities in 2020. Fatalities in road accidents were still higher than those caused by failure to comply with traffic laws, excessive driving behavior, inadequate seat belt use, and disregard for pedestrians and cyclist. Additionally. [6] Oviedo-Trespalacios et al. (2019) assert that driver behavior is an essential element of road safety among road users in the U.S. on the roads. It is necessary to take steps to monitor driver behavior and decrease traffic collisions. Distracted driving, speeding, impaired driving and aggressive behavior are dangerous behaviors that should be checked for. They can cause crashes and fatalities. To reduce injuries and deaths caused by traffic, it is still essential to use education programs as well as traffic law enforcement and public awareness campaigns to promote safer driving practices.

D. Influence of Vehicle Attributes on Road Safety

[2] Marshall (2017) confirms that vehicle attributes significantly influence road safety regarding the severity of crashes and the safety of both driver and passengers. Vehicle attributes such as design, safety features, maintenance, and technological advancements have a role in preventing or exacerbating the likelihood of an accident and the severity of that particular accident. The size and weight of the vehicle increase the possibility of protecting the vehicle occupants from crashes and fatalities during traffic accidents. For example, large vehicles like SUVs and pickup trucks tend to cause fewer fatalities during accidents to their occupants. Vehicle condition and age also have a direct influence on the likelihood of a vehicle causing a traffic accident and the severity of the accident. Older vehicles lacking modern safety features like airbags, electronic stability controls, and advanced driver assistance systems tend to engage in fatal accidents more often. [7] Wu et al. (2013) argue that vehicle maintenance is paramount in ensuring all road users' safety and that the vehicle's age only matters if the vehicle lacks proper maintenance. A well-maintained brake system, tires, lights and steering suspension are vital components that guarantee safe road use for all users. Road accidents resulting in more significant fatalities are mostly linked to vehicle maintenance failure. Even though much research has focused on vehicle automation as the only strategy to reduce vehicle accidents and deaths. Vehicle accidents and fatalities are still relatively high, and vehicle automation needs maintenance to guarantee safe roads. Vehicle attributes play a crucial role in the occurrence and severity of road accidents, albeit significant efforts in vehicle automation seek to reduce human error and accidents. It is worth noting that both maintenance and automation need to happen simultaneously to ensure the safety of all road users.

E. Influence of Environmental Factors on Road Safety

[8] Wang and Zang (2017) concur that environmental conditions have a significant influence on the likelihood of occurrence of an accident and the severity of such accidents. These factors include the weather conditions, the quality of the road surface, lighting and traffic density, which play a primary role in the possibility of an accident. Weather conditions of rain, ice and snow, fog and wind have been found to impair driver visibility and interfere with the road surface's nature, making accidents more likely to occur. Weather conditions impairing the driver's visibility in terms of seeing road signs and markings increases the likelihood of such drivers involving themselves in accidents. Adverse weather conditions like ice, snow and rain change the nature of the road surface, making them slippery and prone to accidents. [9] Dumbaugh et al. (2022) argues that the conditions of the road itself in terms of road surface, lighting visibility and traffic flows contribute to the likelihood of an accident occurring. Poor road surface conditions in terms of potholes, damaged road surfaces and faded road markings and signs make the driving experience difficult and increase the likelihood of drivers making errors. Therefore, a well-maintained road surface is crucial

for efficient driving and helps reduce road accident fatalities with improved driving experience. Also, poor road surface conditions encourage traffic and road congestion, increasing the likelihood of collisions due to dangerous driving and rush hour driving behavior. Environmental conditions have a significant contribution to traffic crashes. In most cases, this can be linked to weather conditions, road surface quality, visibility, traffic density and road design, which influences the risk and severity of an accident.

F. Policy Framework and Road Safety

[**?**] Talbot et al. (2023) define a policy framework as a comprehensive plan to improve road safety in the U.S. It involves policies that shape infrastructure development, driver behavior, vehicle safety standards, and law enforcement. At the federal level, we have the U.S. department of transport with subsidiary agencies like NHTSA and federal highway administration (FHWA), which are mandated to establish national standards, fund infrastructure projects and create regulatory frameworks that enhance the safety of road users. For instance, NHTSA helps implement regulations that determine vehicle safety standards by promoting advanced driver assistance systems that reduce the likelihood of human error. Highway safety improvement programs aim to reduce traffic accidents and fatalities by allocating funds to states to improve roadway infrastructure. The national Roadways Safe strategy targets reducing traffic deaths by incorporating the safe system approach that looks at road safety in a holistic sense of safer people, vehicles, speeds, roads and improved post-crash care. [11] Naumann et al. (2020) underscore that safety campaigns and education have helped shape state-local road safety policies. It has helped shape traffic laws and enforcement by adopting safe speeds and prohibiting distracted distractors by drivers. Impaired driving policies have incorporated 0.008 percent blood alcohol concentration by most states. At the same time, states have invested much in road infrastructure development and maintenance to improve the driving experience and reduce the likelihood of an accident. At the same time, most policy frameworks have emphasized minimizing the consequences of crashes through the safe system approach and technological and innovation policies encouraging vehicle automation to reduce human error. There still exist challenges to the effective implementation of such policies due to funding constraints and lack of uniformity in policy implementation among local states.

G. Interconnectedness of Factors

[3] Job et al. (2022) confirms that road safety in the U.S. is confronted by a complex interplay of human behavior, vehicle attributes, environmental conditions and policy framework. The factors are linked, creating a dynamic system in which changes in one element affect overall road safety outcomes. For example, vehicles with advanced driver assistance systems have helped reduce human error and the likelihood of accident occurrence. Weather conditions also influence drivers' behavior, with adverse weather conditions impairing drivers' vision and driving skills. At the same time, traffic density and road design induce driver stress and aggression that encourages reckless driving and human error. Policy framework policies also influence human behavior, especially speed limits, seat belt laws and drunk driving regulations directly influence driver behavior and help reduce road accidents due to human error. The factors influencing road safety, driver behavior, environmental conditions, vehicle attributes, and policy framework are intertwined elements that require a holistic approach to improve road safety and address traffic fatalities.

H. Conclusion

To conclude, road safety is a measure that ensures the safety of all road users; studying the factors that influence road safety is vital in understanding conditions that promote road safety. Factors such as driver behavior, vehicle attributes, environmental conditions, and policy frameworks have been linked to road safety. These factors have varying degrees of occurrence and severity, as well as traffic crashes and fatalities. It is worth noting that these factors are intertwined to the requirement of a multi-pronged approach that can address all the factors by improving vehicle safety technology, encouraging safer driver behavior, enhancing infrastructure, and promoting comprehensive and effective policies. It is through such holistic approaches aimed at reducing the fatalities emanating from traffic crashes that will help ensure road safety.

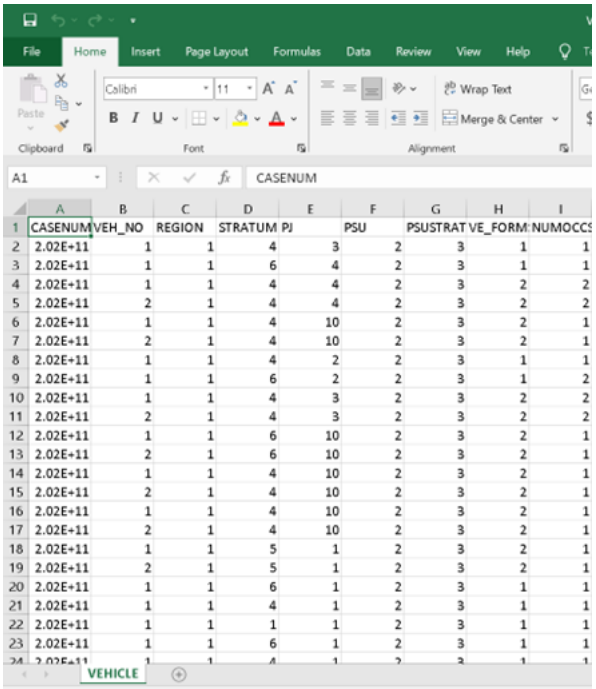
III. DATA

The primary dataset utilized is the GES 2015 Vehicle CSV file, which can be assessed at the following link at the official NHTSA website: NHTSA GES 2015 Vehicle CSV. The target variable is the severity of injury sustained in a crash. This can be categorized into different levels such as No injury, Minor injury, Serious injury and fatality. This target variable is crucial for analyzing the impact of various factors (vehicle characteristics based on our scenario0 on crash outcomes and for predictive models to enhance road safety. https://www.nhtsa.gov/file-downloads?p=nhtsa/downloads/GES/GES15/

IV. METHODOLOGY

A. Technology Used

- Google Drive for Data Storage: The dataset was stored on Google Drive, ensuring easy access and collaboration throughout the analysis process.
- Python: Python was used as the main programming language for data processing, cleaning, and analysis. Key libraries such as Pandas, NumPy, and Scikit-learn were employed for various tasks like data manipulation, model building, and evaluation.
- Google Colab: The cloud-based environment of Google Colab was utilized for Python code execution and visualization, with the added benefit of using Google Drive to store data.
- Handling Incomplete Rows: Rows with missing or incomplete data were identified and removed to maintain the quality and reliability of the dataset.



CASENUM	VEH_NO	REGION	STRATUM	PJ	PSU	PSUSTRAT	VE_FORM	NUMOCCS
2.02E+11	1	1	4	3	2	3	1	1
2.02E+11	1	1	6	4	2	3	1	1
2.02E+11	1	1	4	4	2	3	2	2
2.02E+11	2	1	4	4	2	3	2	2
2.02E+11	1	1	4	10	2	3	2	1
2.02E+11	2	1	4	10	2	3	2	1
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2.02E+11	2	1	5	1	2	3	2	1
2.02E+11	1	1	6	1	2	3	1	1
2.02E+11	1	1	4	1	2	3	1	1
2.02E+11	1	1	1	1	2	3	1	1
2.02E+11	1	1	6	1	2	3	1	1

Fig. 1. Sample visual representation of the CSV data

B. Data Cleaning

Addressing Missing Values: Missing values were handled by either imputing with mean, median, or mode values where appropriate or removing rows where the missing data was too extensive to impute reliably.

The identification and treatment of outliers in the dataset through statistical techniques (such as Z-squares and IQR) involved the appropriate action of eliminating or altering them, depending on their potential contribution to the analysis.

C. Feature engineering

Domain Knowledge and exploratory data analysis were used to select relevant features for predicting the target variable, maximum injury severity. The features comprised accident details, vehicle characteristics, and environmental conditions.

a) Defining the Target Variable: : The target variable, Maximum Injury Severity, was clearly identified. This variable, representing the highest injury level in an accident (e.g., minor, moderate, severe, or fatal), became the focus of the predictive model.

b) Data Splitting:: The dataset was split into training and testing sets to evaluate the performance of the predictive models. An 80-20 split was used, with 80 percent of the data allocated for training and 20 percent for testing. This allowed for model training and unbiased evaluation.

V. RESULTS

Detailed information on road safety in the United States is available through the National Highway Traffic Safety Administration's (NHTSA) General Estimates System (GES) data from 2015. We can derive insights into a variety of factors

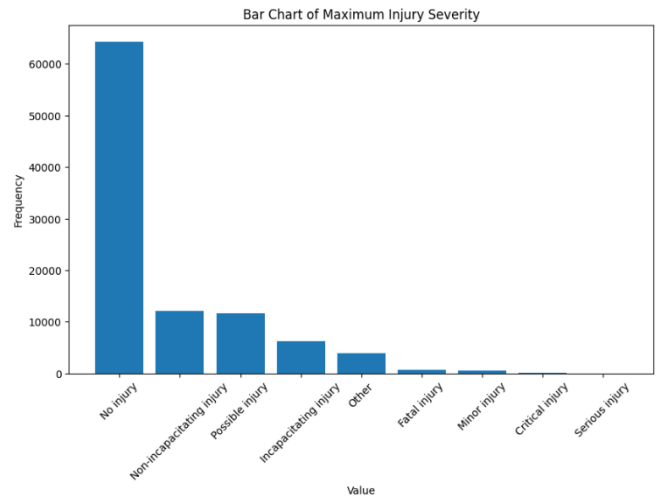

Fig. 2. Bar chart of maximum injury severity.

that impact the frequency and severity of traffic accidents using EDA methods. Listed below are the main findings:

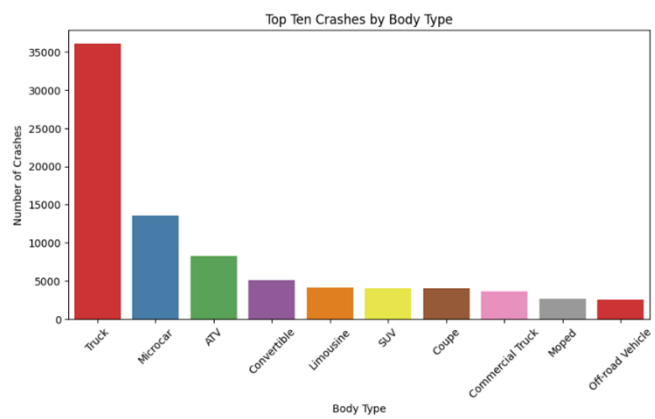

Fig. 3. Car crash by body Type.

a) Prevalent Body Types in Crashes: : Trucks represent the highest number of crashes, followed by microcars and ATVs. Notably, larger vehicles such as SUVs, convertibles, and limousines also appear frequently in crash data. While mopeds and off-road vehicles account for a smaller share, their involvement in accidents remains noteworthy.

b) Vehicle Makes in Accidents: : The Bricklin and Daihatsu brands emerge as the most frequently involved in crashes, followed by popular makes like Mazda, Isuzu, and Infiniti. Interestingly, Dodge and Mitsubishi also show up among the top 10, alongside fewer common brands, underscoring the diversity of vehicle makes involved in incidents.

c) Injury Severity: : The dataset is dominated by non-injury accidents, with non–invalid injuries and possible injuries being the next most common. While serious injuries do occur, fatal ones are reported as well, and the dataset also includes both minor and critical injuries, with serious injury being extremely rare.

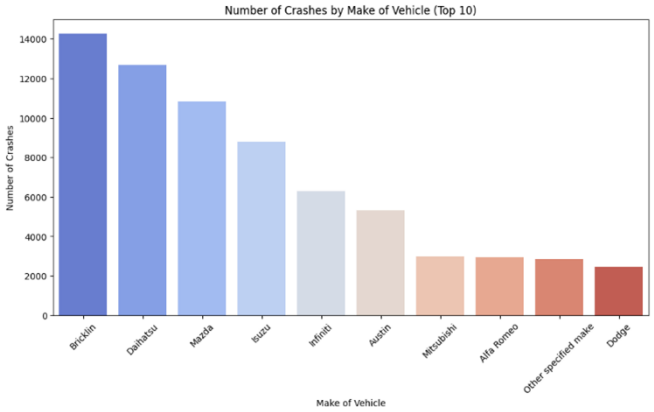

Fig. 4. Numbers of crashes by make of vehicle.

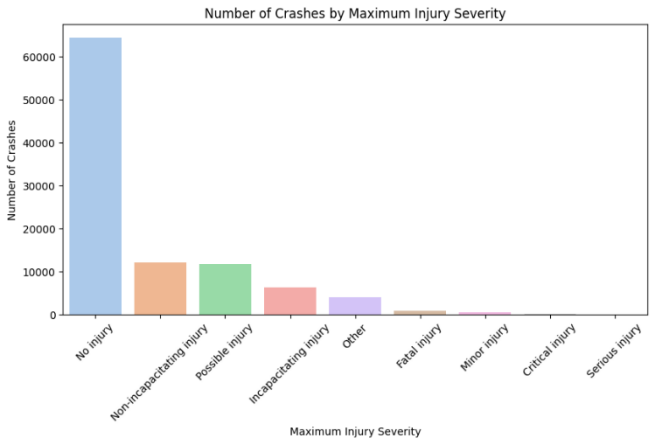

Fig. 5. Number of accidents by maximum injury severity.

- Common First Harmful Events: Unexploded gas, animal-caused crashes, traffic signs, and car collisions are the primary causes of accidents. The number of crashes is also influenced by other factors, including driver distractions and incidents of road rage.
- Manner of Collision: Common types of collisions include head-on collision, bicycle collision and incidents involving no collision (e.g. single-vehicle crash).[Note]. Rear-end crashes and single-vehicle collisions are among the other significant factors. While less common, roll-over accidents are still a significant safety concern.
- Alcohol Involvement: Alcohol is a significant factor in crashes, with the majority of incidents being classified as yes on the alcohol indicator, but some variables are uncertain and no alcohol categories also make up 80 percent of the data. Despite some uncertainty in the reporting, alcohol is still a significant factor in road safety concerns.
- Pre-Crash Actions: Most drivers react and decide before a crash by decelerating and avoiding obstacles. This is the role of response decisions. What should you do? Other significant maneuvers involve approaching intersections, stopping and starting to reverse. The outcomes demonstrate that proactive actions, such as enhancing driver awareness and providing brake support for safer driving could help reduce accidents.

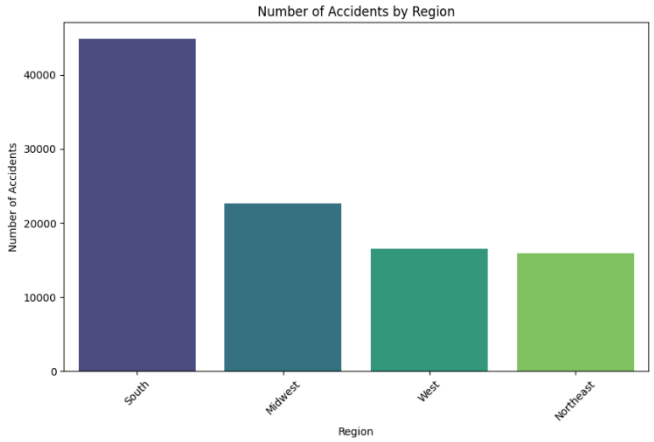

Fig. 6. Number of accidents by region.

- Accident Types: Accidents resulting from distraction from other passengers and weather-related events make up a significant portion of the crash data. Behavioral and environmental factors play a role in road safety, with road rage (incendiaries), blind spot collisions, and wrong-way driving being significant examples.
- Geographic Patterns: The majority of accidents occur in the South, while most occur throughout the Midwest, West, and Northeast. Variations in weather, traffic patterns, and driver behavior are reflected in regional differences across the country. The South has the highest accident rates among all regions.

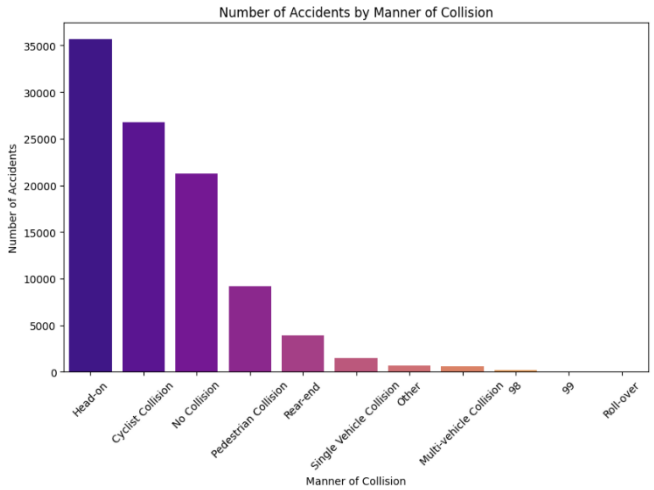

Fig. 7. Number of accidents by manner of collisions.

- Unit Types and Injury Severity: The type of vehicle most commonly involved in crashes is passenger car, and there are the highest percentage of injuries. Nonetheless, some

vehicle types, such as motorcycles and large trucks have been associated with more serious injuries; therefore, safety measures must be targeted.).

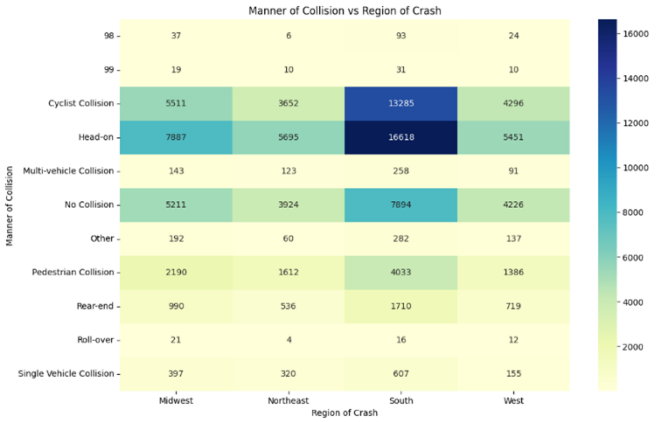

Fig. 8. Contingency plot of manner of collision vs region of crash.

- Contingency Table Insights: The correlation between the manner of collision and crash area indicates regional patterns. Pedestrian collisions are more common in the South, while rear-end collision occurs more often in Midwest. The patterns suggest that specific safety measures may be more effective in certain areas based on the type of accident that occurs frequently. Data collected by the NHTSA GES 2015 provide a detailed assessment of road safety, including how vehicle type and pre-crash actions affect accidents as well as whether alcohol consumption or regional differences influence them. By addressing these trends, improving driver education, and targeting high-risk areas and vehicle types to improve road safety, we can achieve significant improvements in road infrastructure.

A. *Exploratory Data Analysis*

What is the relationship between vehicle characteristics (make, model, year, body type) and the severity of crash outcomes?

Crash severity and its make, model, year, and body type are important factors that impact vehicle characteristics. What are they? Alfa Romeo, a luxury and high-performance car brand, has exhibited heightened risk reduction with its higher rate of no injury incidents. This suggests that more advanced safety features or vehicle design may be responsible for reducing the severity of injuries. The prevalence of fatal and critical injuries in less common or outdated vehicle models such as ”Other specified make” may be due to the absence of modern safety features, crashes, or design improvements.

How do specific vehicle attributes (e.g., safety features, weight, size) influence the likelihood and severity of collisions?

Vehicle safety characteristics, vehicle weight, and size play a significant role in determining crash severity and likelihood. The size and force of collisions with larger, heavier vehicles

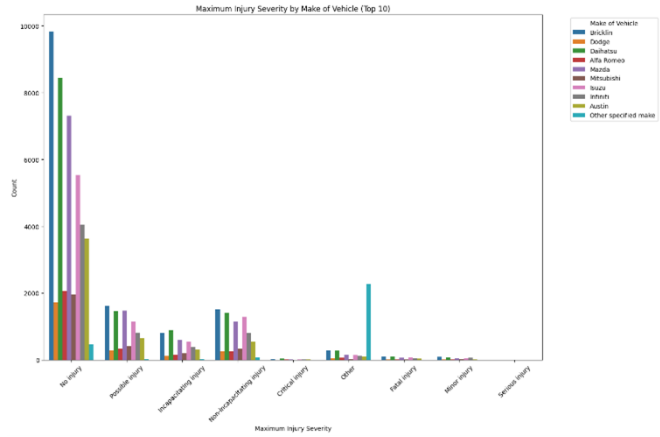

Fig. 9. Maximum injury severity by make of vehicle.

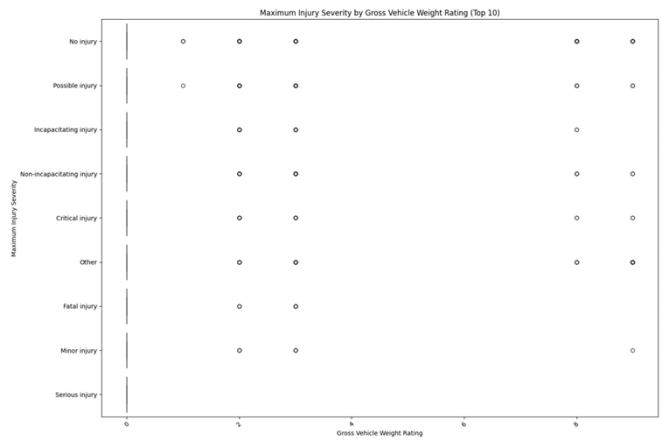

Fig. 10. Maximum injury severity by gross vehicle weight ratings.

like trucks can increase the likelihood of serious accidents, such as blind spot collision or wrong-way driving. In contrast, smaller vehicles such as ATVs or microcars are more prone to injury, with weather-related accidents and distractions being more common. Enhanced safety features in modern vehicles may be effective against some of the worst crashes caused by human error, as evidenced by the lower number of injuries reported from newer models.

What patterns can be identified in crash occurrences based on vehicle types involved (e.g., passenger cars vs. trucks) and their characteristics?

Different vehicle types exhibit distinct patterns in crashes. While passenger cars are involved in most rear-end collisions, trucking is associated with more serious accidents, such as blind spot collision and wrong-way driving. Heavy collisions are often caused by larger vehicles, which are also characterized by their size and design.

How does the age of a vehicle correlate with the frequency and severity of accidents?

Vehicle age appears to be linked to accident frequency/severity, which is also related. The absence of modern safety features or aging vehicles can lead to higher accident

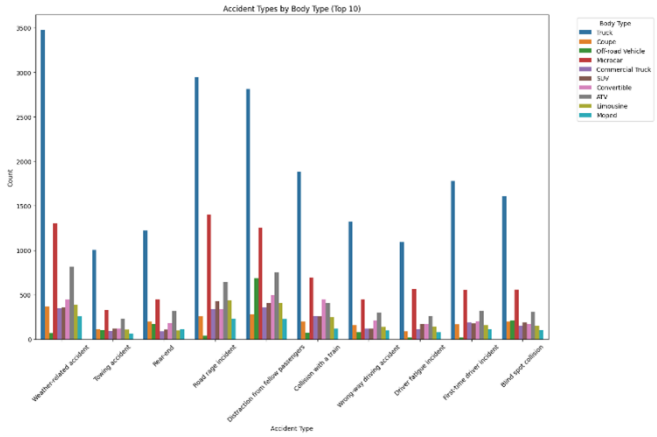

Fig. 11. Accident Type by body Type

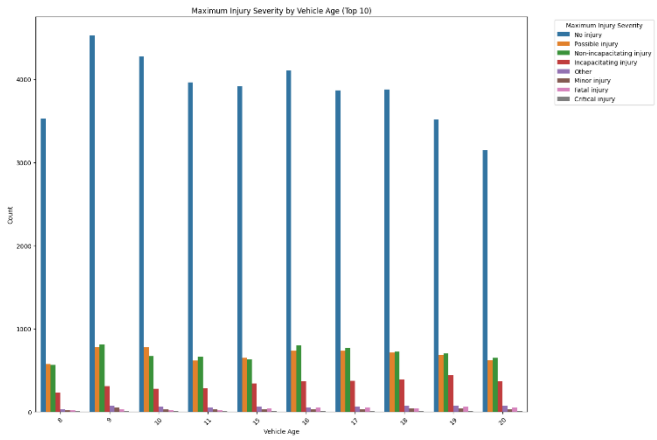

Fig. 12. Maximum Injury Severity by Vehicle Age.

rates. The average age of vehicles involved in crashes is roughly 14 years, and older vehicles are more likely to have serious injuries or fatalities. As a result, the likelihood of accidents and the severity of crash incidents can rise due to vehicle age and potential wear and tear.

Hypothesis Testing Result and Insight

The Mann-Whitney U test reveals a statistically significant difference in the severity of injuries between newer and older vehicles (p-value = 0.0000). For newer vehicles (those 10 years old or younger), the severity of injuries tends to be lower, with a median injury severity score (MAX VSEV) of 0.00, suggesting that these vehicles are more likely to result in minor or no injuries. In contrast, older vehicles (over 10 years) have a higher median injury severity score of 0.00, but with broader ranges of injury intensity, indicated by an IQR of 1.00. The implication is that accidents involving older vehicles are more likely to result in severe injuries. Newer vehicles are likely to have more advanced safety features, while older vehicles may not have these systems, leading to more severe crashes.

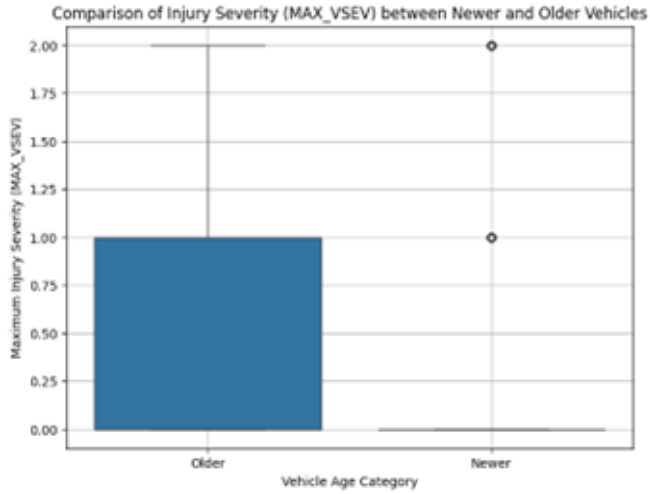

Fig. 13. Comparison of injury severity between newer and older vehicles.

B. Machine learning Results

To predict the severity of injuries classified as severe and non-severe in vehicle accidents, various machine learning models were used in this analysis. The aim of the modelling process was to identify the most appropriate model for predicting injury severity by considering various vehicle features. We utilized four models, namely Random Forest, Decision Tree, Logistic Regression, and a Neural Network Model (MLP), to achieve this. The evaluation of each model was based on essential metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Model Comparison Results

Here is a summary of the evaluation results for the models tested:

TABLE I
MODEL PERFORMANCE METRICS

Model	Accuracy	ROC-AUC	Precision (Severe)	Recall (Severe)	F1-Score (Severe)
Random Forest	0.7342	0.6382	0.4079	0.1604	0.2303
Decision Tree	0.6556	0.5448	0.3138	0.3284	0.3209
Logistic Regression	0.7521	0.5956	0.4286	0.0009	0.0018

Based on the results, Logistic Regression performed the best in terms of accuracy (75.21 percent) but struggled significantly with detecting severe injuries, as indicated by its very low recall and F1-score for the Severe class. Random Forest provided a balanced performance, with decent accuracy (73.42 percent) and a more robust ability to identify severe injuries, although it still had room for improvement. Based on these metrics, Random Forest emerged as the most reliable model for predicting injury severity, particularly when it comes to identifying non-severe injuries.

VI. CONCLUSION

This study underscores the challenges faced in ensuring road safety throughout America, even with substantial tech-

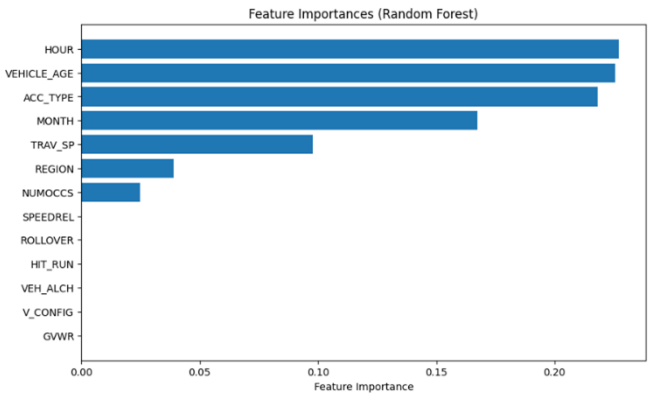

Fig. 14. Feature importance for random Forest.

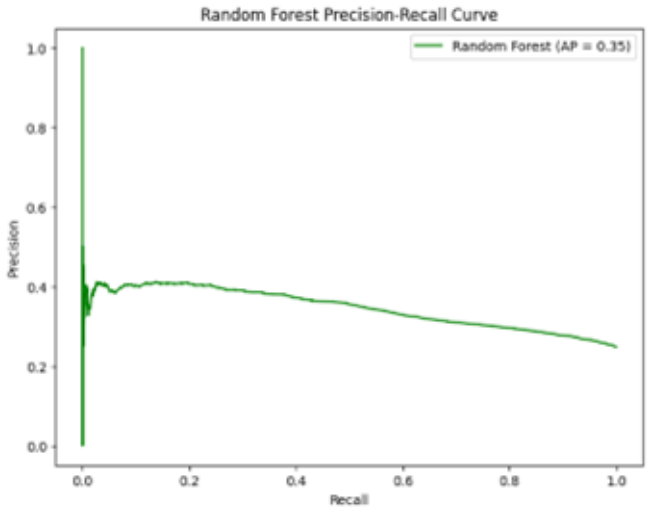

Fig. 15. Random Forest precision recall curve..

nological and infrastructure improvements. Driver behavior, vehicle features, environmental conditions, and infrastructural factors are all factors that can impact crash outcomes. By focusing on unsafe driving behaviors, investing in vehicle safety technology, and building infrastructure, the outcomes suggest that there are many ways to achieve greater road safety. Not only does this demonstrate that using machine learning models to predict severity of crashes, it also shows that data-driven approaches can be used to identify at-risk vehicles and scenarios but further research needs ongoing to improve these methods.

VII. RECOMMENDATIONS

Several practical measures have been identified by us to improve road safety. It is important to conduct targeted public awareness campaigns on the hazards of distracted and impaired driving to minimize human error. In addition, it is crucial for policy makers to create laws that encourage the inclusion of safety features in vehicles to ensure they meet current safety requirements. It is especially important for older vehicles. Enhanced infrastructure, such as better road surface maintenance and safer traffic environments, is essential in reducing the risk of major accidents. Additionally. Advances in predictive technology have enabled authorities to anticipate and prevent traffic accidents more effectively.

VIII. FUTURE WORKS

The dataset's size should be broadened to encompass more recent years for future research on road safety trends and the impact of new vehicle technologies. Predictive modeling of traffic incidents can be improved by incorporating real-time data analytics. Future studies could examine how new trends, such as self-driving cars, affect safety features and whether they have the potential to change road safety. Also, we can see how much more effectively machine learning outputs can be integrated with policy recommendations to ensure national road safety.

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